

Reference-Component Extraction from Hebrew-Aramaic Responsa under Low-Resource and Imbalanced Conditions

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Abstract

This paper presents a controlled study of reference-component extraction from Hebrew-Aramaic Jewish Responsa literature. The task is formulated as token-level named entity recognition over a small annotated corpus of 300 sentences and approximately 6,800 tokens, with 898 entity tokens and a dominant non-entity class. Four Hebrew transformer encoders are evaluated: DictaBERT, BEREL 3.0, HeRo, and AlephBERT-Gimmel. The study compares sentence-level split strategies, targeted masked-language-model augmentation, regular one-stage NER, a cascaded three-step pipeline, confidence-based fusion, and SVM-based fusion. All comparisons use 20 seed-aligned runs and paired Wilcoxon signed-rank tests. The strongest robust setting is BEREL 3.0 with Multilabel Stratified splitting, augmentation, and SVM Fusion, reaching 0.852 mean F1. A conservative non-router setting, BEREL 3.0 with Multilabel Stratified splitting, augmentation, and Confidence Fusion, reaches 0.848 mean F1. The results show that in small specialized Hebrew-Aramaic corpora, data design and inference design are as important as the choice of pretrained encoder.

Keywords: named entity recognition; Hebrew NLP; Responsa literature; low-resource learning; data augmentation; citation extraction; transformer models.

1 Introduction

Jewish Responsa literature contains many references to earlier authors, books, chapters, and sections. Extracting these references is useful for digital humanities, legal history, and citation-network analysis (Satlow and Sperling, 2022; Ben-Gigi et al., 2025). This paper focuses on a smaller technical subtask: token-level extraction of the reference components themselves.

The task is challenging because the data is small and imbalanced, the text is Hebrew-Aramaic, and references are often abbreviated or only partially structured. A standard transformer fine-tuning pipeline can produce unstable conclusions in this setting because a single random split may underrepresent rare labels. Therefore, this study asks whether better splitting, targeted augmentation, and fused inference can improve low-resource reference-component NER.

2 Related Work

Reference extraction from rabbinic literature is connected to earlier work on biblical quotation identification, dating of rabbinic texts, and citation-network analysis (HaCohen-Kerner et al., 2010; Moghaz et al., 2019; Satlow and Sperling, 2022). The closest recent study is Ben-Gigi et al. (2025), which automatically constructs citation networks from medieval Jewish Responsa

literature. The present study is complementary: it does not claim a direct benchmark comparison, but analyzes how to train and evaluate reference-component NER when only a small imbalanced annotated dataset is available.

The encoder comparison uses recent Hebrew language models: AlephBERT (Seker et al., 2022), AlephBERT-Gimmel (Gueta et al., 2023), HeRo (Shalumov and Haskey, 2023), DictaBERT (Shmidman et al., 2023), and BEREL for rabbinic-encoded language (Shmidman et al., 2022). The method design is also related to low-resource NER, sequence constraints, data augmentation, and classifier fusion (Li et al., 2022; Nguyen et al., 2023; Zhou et al., 2022; Florian et al., 2003; Awasthy et al., 2020).

3 Data and Task

The final corpus contains 300 annotated sentences, about 6,800 tokens, and 898 entity tokens. Most tokens are labeled 0. The entity schema includes author, book, chapter, section, subchapter, and prefix markers that introduce these components.

Table 1: Entity distribution in the annotated corpus.

Entity	Tokens	Percent
AUT	308	34.30
BOK	180	20.04
CHP	120	13.36
SCHP	58	6.46
SEC	75	8.35
pCHP	81	9.02
pSCHP	54	6.01
pSEC	22	2.45

Preprocessing experiments showed that standard tokenization was not sufficient for the corpus because abbreviations and punctuation marks are important for reference boundaries. The final preprocessing used a Hebrew-oriented tokenizer that preserves abbreviations and separates full-stop punctuation from partial-stop punctuation. Abbreviation expansion was evaluated manually on a sample and reached 72.77 percent correct expansions.

4 Methods

The study compares three sentence-level train-test splits. The baseline random split assigns about 70 percent of sentences to training and 30 percent to evaluation. The label-aware greedy split tries to preserve non-0 entity coverage in the training set. The multilabel stratified split treats each sentence as a set of labels and assigns sentences to preserve label proportions, following multilabel stratification (Sechidis et al., 2011).

Targeted augmentation is applied only to the training set. For label ℓ , the deficit is

$$\Delta(\ell) = f_{max} - f(\ell),$$

where f_{max} is the largest entity-label frequency in the training set. The number of generated examples is proportional to this deficit. New sentences are generated through masked entity replacement, preserving the original BIO structure.

The four reported NER methods are:

1. **Regular NER**: a transformer token classifier that directly predicts full BIO labels.
2. **Cascaded 3-step**: a decomposed model that predicts entity membership, boundary position, and entity type, followed by BIO consistency repair.
3. **Confidence Fusion**: a token-level fusion rule that keeps agreement between Regular NER and Cascaded 3-step and resolves disagreement by confidence.
4. **SVM Fusion**: a learned router that decides which source to trust on disagreement tokens.

The SVM Fusion result should be read cautiously because the router was not trained on a fully independent held-out routing set. It is useful as an exploratory estimate of the value of routing, while Confidence Fusion is the more conservative fusion result.

Evaluation uses entity-level F1. All experiments are repeated across 20 seed-aligned runs, and paired Wilcoxon signed-rank tests are applied to matched runs.

5 Results

Table 2 presents the main Multilabel Stratified results. Augmentation raises the performance level across all models. BEREL 3.0 becomes the strongest robust model after augmentation, which is consistent with its rabbinic-domain orientation.

Table 2: Main Multilabel Stratified results. Values are mean entity-level F1 $\pm$ standard deviation over 20 paired seeds.

Training data	Method	DictaBERT	BEREL 3.0	HeRo	AlephBERT-Gimmel
Original	Regular NER	0.738 $\pm$ 0.000	0.731 $\pm$ 0.006	0.769 $\pm$ 0.000	0.627 $\pm$ 0.004
Original	Cascaded 3-step	0.764 $\pm$ 0.024	0.639 $\pm$ 0.147	0.750 $\pm$ 0.022	0.728 $\pm$ 0.031
Original	Confidence Fusion	0.745 $\pm$ 0.013	0.739 $\pm$ 0.012	0.776 $\pm$ 0.006	0.702 $\pm$ 0.020
Original	SVM Fusion	0.807 $\pm$ 0.010	0.795 $\pm$ 0.030	0.805 $\pm$ 0.011	0.766 $\pm$ 0.015
Augmented	Regular NER	0.808 $\pm$ 0.010	0.847 $\pm$ 0.008	0.821 $\pm$ 0.009	0.834 $\pm$ 0.006
Augmented	Cascaded 3-step	0.828 $\pm$ 0.009	0.846 $\pm$ 0.008	0.825 $\pm$ 0.014	0.839 $\pm$ 0.010
Augmented	Confidence Fusion	0.814 $\pm$ 0.012	0.848 $\pm$ 0.008	0.823 $\pm$ 0.009	0.832 $\pm$ 0.008
Augmented	SVM Fusion	0.836 $\pm$ 0.008	0.852 $\pm$ 0.008	0.841 $\pm$ 0.010	0.847 $\pm$ 0.008

The highest numerical mean F1 in the workbook is 0.857 for BEREL 3.0 with baseline random split, augmentation, and SVM Fusion. However, the main robust result is 0.852 for BEREL 3.0 with Multilabel Stratified split, augmentation, and SVM Fusion. This distinction matters: the random split gives a slightly higher score in this one condition, but the stratified split is more reliable for rare-label coverage.

Table 3: Best mean F1 configuration for each model among the reported methods.

Model	Best method	Split/data condition	F1
DictaBERT	SVM Fusion	Label-aware greedy + augmentation	0.841 $\pm$ 0.013
BEREL 3.0	SVM Fusion	Baseline random + augmentation	0.857 $\pm$ 0.008
HeRo	SVM Fusion	Multilabel stratified + augmentation	0.841 $\pm$ 0.010
AlephBERT-Gimmel	SVM Fusion	Multilabel stratified + augmentation	0.847 $\pm$ 0.008

Table 4 summarizes the paired Wilcoxon tests. Augmentation improves all 48 matched comparisons. Multilabel stratification is especially strong in the augmented setting. SVM Fusion improves over the other reported methods in nearly all comparisons, but its router limitation should be considered when judging deployable performance.

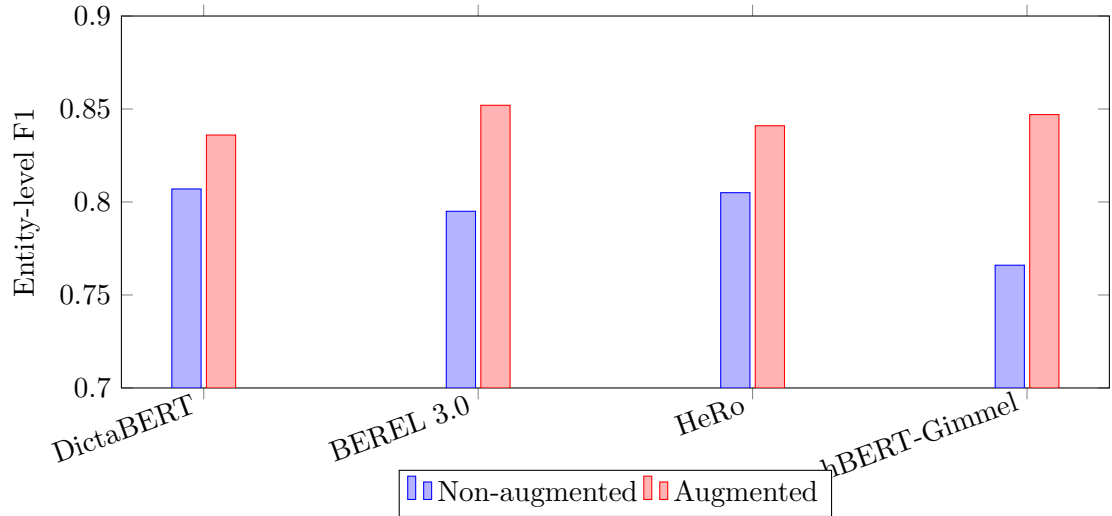


Figure 1: SVM Fusion with Multilabel Stratified split, before and after augmentation.

Table 4: Summary of paired Wilcoxon signed-rank tests over seed-aligned runs. Positive means that the first item in the contrast improved F1.

Contrast	Setting	Positive	Significant	Mean Δ F1	Largest p
Augmentation vs. same split without augmentation	All models, methods, splits	48/48	48/48	+0.085	3.9e-04
Label-aware greedy vs. baseline random	Non-augmented	14/16	10/16	+0.021	0.898
Multilabel stratified vs. baseline random	Non-augmented	15/16	13/16	+0.024	0.927
Label-aware greedy vs. baseline random	Augmented	11/16	13/16	+0.009	0.430
Multilabel stratified vs. baseline random	Augmented	15/16	16/16	+0.028	0.033
SVM Fusion vs. Regular NER	Non-augmented	12/12	12/12	+0.073	8.8e-05
SVM Fusion vs. Cascaded 3-step	Non-augmented	12/12	12/12	+0.071	2.6e-04
SVM Fusion vs. Confidence Fusion	Non-augmented	12/12	12/12	+0.057	9.5e-06
SVM Fusion vs. Regular NER	Augmented	12/12	12/12	+0.025	0.019
SVM Fusion vs. Cascaded 3-step	Augmented	12/12	12/12	+0.023	0.024
SVM Fusion vs. Confidence Fusion	Augmented	12/12	11/12	+0.025	0.076

6 Discussion

The results show that the final performance is not determined by the pretrained model alone. The same encoder can behave differently depending on the sentence split, the use of augmentation, and the inference pipeline. This is especially important in small corpora, where random train-test splits can be misleading.

The comparison with Ben-Gigi et al. (2025) is best understood as complementary. Their study focuses on large-scale Responsa citation-network construction. This study focuses on controlled low-resource methodology: how to split, augment, model, fuse, and test when annotation is scarce. The added value is therefore methodological rather than a direct benchmark improvement.

A practical recommendation is to avoid drawing conclusions from a single random split in this domain. A better design is to use multilabel sentence-level stratification, keep augmentation strictly within the training set, and report paired multi-seed significance tests. For deployment, the SVM router should be retrained with a separate routing-development set.

7 Conclusion

This paper presented a controlled study of Hebrew-Aramaic Responsa reference-component extraction as low-resource NER. The strongest robust setting uses BEREL 3.0, Multilabel Stratified splitting, targeted augmentation, and SVM Fusion, reaching 0.852 mean F1 over 20 paired seeds. A conservative non-router alternative, BEREL 3.0 with Multilabel Stratified splitting, augmentation, and Confidence Fusion, reaches 0.848. The main conclusion is that data preparation and inference design can be as important as model selection in small specialized Hebrew-Aramaic corpora.

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